Carnegie Mellon University
Computer Science Academy

Design and Development Guide

Before beginning this project, be sure to review the rubric to ensure you include all the required elements.

Description

Describe your program. What do you envision your app will do?

Note: At this step, do not be concerned about how to do it, just focus on what you envision.

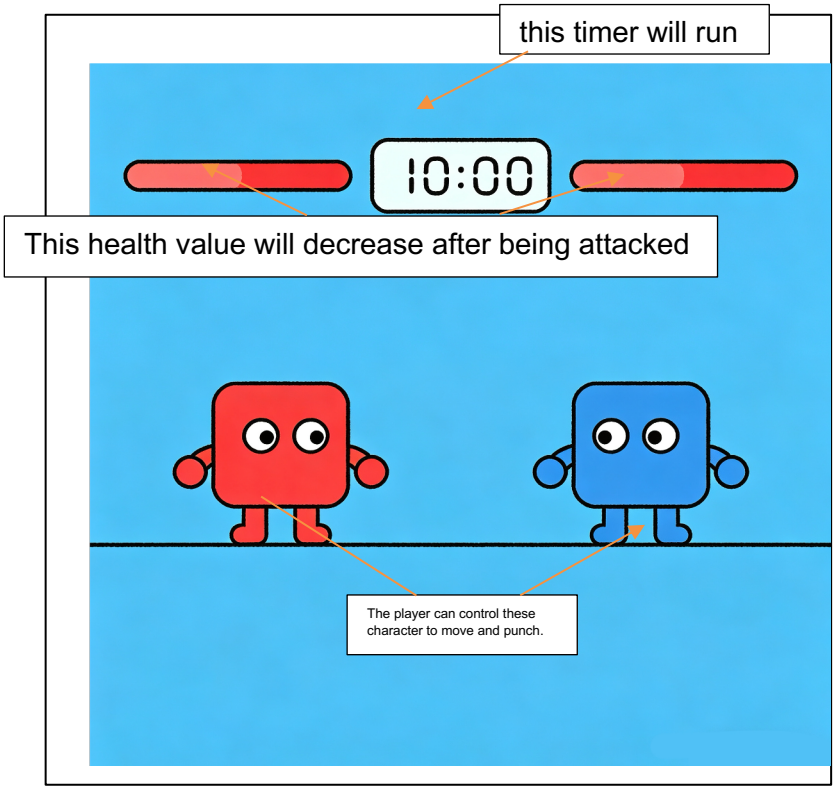
I want to create a 2.5D fighting game, similar to the game "Mugen". Players can move left and right using the 'a' and 'd' keys, and can also make small up-and-down movements using the 'w' and 's' keys. Pressing the spacebar allows them to punch and attack. If the fist hits an enemy, it can reduce the enemy's health. The player whose health runs out loses.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?

(0,0)a



(400,400)

Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

Concept	Purpose
OnKeyHold	Used for interaction between players and game characters， Used to control the player's movement up, down, left or right
OnkeyPress	Used to control the character's attack action （punch）
Fist1.dx=...	This is the core content of this unit. I used it to create the reciprocating motion of the fist, achieving an effect of punching
onStep	Used to implement all animations, including the movement of characters, the action of punching, as well as the changes of timers and health bars

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

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For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
  I encountered some program bugs and technical difficulties, but they were all resolved in the end. For example, when I was writing the code for the punching action for each character. When I press “space” both of the characters punches instead of only player 1 punches.
- Attempted Solutions: What did you try in order to solve the problem?
  Because I set it so that player1’s punching was done by pressing the "space", I then used the command + f to check all the code related to the space bar. To my surprise, it turned out to be the same for all cases. This really confused me. It was not until the teacher's words reminded me of the problem that I realized it.
- Final Solution: What was the final solution that fixed the issue?
  This time, I didn't just look at the line of code related to "space", but examined the entire section. Eventually, I discovered that I accidentally used the "if in keys" statement in the "OnKeyPress" section.

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
  Because I considered that if this was an attack on the same plane, the game would lose its fun. So I decided to allow players to move slightly up and down, so that they could attack.
  real characters and virtual characters.
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
  In fact, there isn't.
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?
  I need to make the stage 3D (I use slashes to represent it) and add additional keys to control the character's movement up and down.

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose

Line()	5-11	Draw a line between the two points I have specified.	I created a 3D-like visual effect by using multiple angled lines of different angles.
def moveleft():	52	When the player presses 'a', the x-coordinate of the character decreases.	I defined these functions, and then used onKeyHold to activate them, thereby achieving the effect of allowing the player to control the character's movement.
def onStep():	167	It will automatically trigger multiple times per second.	Simulated the animation of the fist being thrown, as well as the animation of the blood quantity decreasing
Dx	145 and 193	A custum property of a object	By changing the value of dx, I made the fist move away from the character after pressing "space", and it would reverse direction after moving 50 units, then pause when it returned to the player's side.

Future goals

Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
  If possible, I would add feedback when the character is attacked, such as making the character's body turn red or etc. SHowing that he has been attacked. However, this would involve creating a completely new character with red skin, hiding it initially and then revealing it when attacked. Since it was too complicated, I didn't do it.
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
  Overall, I followed my original plan. The only change I made was to increase the vertical movement space for the characters, which was very beneficial and made the game more enjoyable.
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
  I did utilize a lot of resources, such as the assistance from teachers and classmates. I also tried to search for similar games online to draw inspiration from them.
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?
  I did made efficient use of my time. In each class, I was concentrating on programming. Sometimes I would listen to music, which helped me work more effectively. I will also turn on the "Do Not Disturb" mode on my computer. In that case, I will be more focused.

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.